BT4301 – Business Analytics Solutions Development and Deployment

AY 2025/26 Semester 2

Group Project

Overview of Project

You are required to form a 5-member group to undertake a project on DataOps and MLOps. If the final cohort enrolment size is not a multiple of 5, one or more groups may be allowed to have either 4 or 6 members.

The project is broadly divided into two different phases – **ideation phase** and **implementation phase**.

- **Ideation Phase** – Conceptualise an analytics project within the context of a business organisation in a specific industry that deliver one or more tangible business values. Your group may work on any business industry/sector. The analytics project must include **at least one machine learning (ML) model** as the core deliverable. Other supporting output such as data dashboards or reports that are relevant to the ML model(s) may also be included as part of the project deliverable. Your group may also consider implementing a scaled-down software application demonstrating the integration and deployment of the ML model(s).

  The analytics project should also involve at least one full-fledged dataset with a substantial number of variables and observations. Your group may choose to work on structured data or unstructured data (text or images) depending on the collective expertise of group members. There is no distinction in the evaluation process between the use of structured and unstructured data.

- **Implementation Phase** – Implement the proposed analytics project using the **Scrum agile methodology** in conjunction with appropriate **DevOps, DataOps and MLOps** processes. In particular, the DataOps processes should be integrated with the MLOps processes such that the training and validation data are seamlessly ingested into model development and productionisation.

  As part of project documentation, your group should cater to the management plan of the entire project. But the actual project duration for evidencing Scrum management is only two sprints of one to two weeks each. The first sprint should involve DataOps tasks, and the second sprint should involve MLOps tasks.

  As part of project implementation, your group is required to use suitable toolchains to support DataOps and MLOps processes. The minimal requirements are the use of proper orchestration tools and a server-based relational database system (SQLite is

not allowed). The use of other tools is optional, e.g., data ingestion, integration, transformation, etc.

Please organise your group members to take on different roles and responsibilities such as Scrum Master, Product Owner and Team Members. There is no distinction in the evaluation process between different roles and responsibilities.

Your group is required to submit the final deliverable consisting of a written report, datasets and source code in Week 13. Please reach out to the lecturer if you group requires any project consultation or any other assistance.

Your group will demonstrate the project deliverable via two different but complementary modes:

- Peer learning and sharing – In-person presentation on campus during the last lecture in Week 13.
- Full-demonstration – Prerecorded video to be submitted by Tuesday of reading week, i.e., a two-day buffer is provided for the completion of the video recording after the submission deadline.

Group Registration

Please register your group via Canvas – People/Groups. Do ensure that all members are registered in the same group.

You can use the Canvas Discussions tool > Recruitment of Project Group Members for members recruitment. Please complete the group formation latest by the end of Week 4, i.e., **Sunday, 8 February, 11:59 pm**.

Overall Assessment Criteria

The overall assessment criteria are listed below.

Formulation of Analytics Project • Problem statement • Business value(s) • Solution statement	**5%**
Agile Analytics Project Management with Scrum • Overall project planning. • Detailed sprint planning for two sprints of two weeks each. • One sprint for DataOps • One sprint for MLOps	**10%**
DataOps • Dataset construction • Dataset ingestion, integration, cleansing, transformation, lineage, exploratory analysis, etc. • Automation and orchestration of DataOps process with suitable toolchain	**15%**

MLOps • Model development – Data exploration, feature engineering and selection, experimentation, model evaluation and comparison, version management and reproducibility, etc. • Preparing for production • Deploying to production • Monitoring and feedback loop • Model governance	**15%**
DevOps • Model integration into software application. • Scaled-down version of software application with minimal user interface and functionalities is acceptable.	**5%**
Total	**50%**

General Project Schedule

S/N	Week	Date	Activity	Remark
1	4	8 Feb	Project group registration	
2	6	19 Feb	Release of project specification	
3	6 to 13	-	Project tasks and ad-hoc consultations	
4	13	16 Apr	Peer learning and sharing	Done in-class during last lecture
5	13	19 Apr	Submission of project deliverable Submission of peer review	Done via Canvas by 11:59 pm
6	Reading	21 Apr	Submission of full demonstration video	Done via Canvas by 11:59 pm

Formulation of Analytics Project

State the organisational context by highlighting the business nature of the organisation together with the industry/sector it is operating in. State the business question and provide a description of the relevant business challenge/problem/opportunity that can be addressed with analytics. Explain the associated business value(s). Then identify the affected business processes and the relevant business units and/or functional departments.

State the analytics solution that your group is implementing to address the business question. The core deliverable must include at least one ML model. Explain how the affected business processes should be redesigned to seamlessly integrate the ML model(s) to deliver the expected business values preferably with the involvement of automation via integration with a software application.

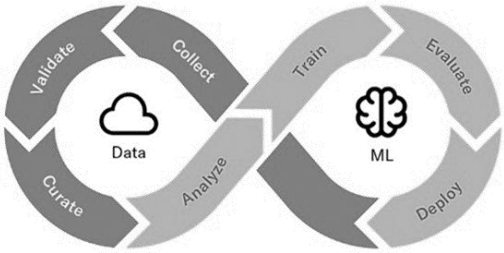
Your group is encouraged to refer to the Target's pregnancy predictor case study discussed in Lecture 01 for an example of a real-world predictive analytics project that delivers tangible business values.

Agile Analytics Project Management with Scrum

Perform overall project planning using the Scrum methodology and outline the project backlog and sprints that are required to deliver the solution in an iterative and incremental manner. The planning should cover the entire life cycle of the project beginning with

business understanding and leading to model deployment, model maintenance and demonstration of business values. In typical agile analytics style, the project is expected to be undertaken on an ongoing basis with continuous refinement of the models. For the purpose of this project, the planning may stop after a reasonable number of sprints leading to the delivery of the initial versions of the model.

The planning should cover both DataOps and MLOps processes with both set of processes running concurrently and integrating seamlessly, exemplifying the essence of agile analytics.



Your group is also required to perform detailed sprint planning for two sprints of one to two weeks each. One sprint should focus on DataOps and another sprint focusing on MLOps. Each detailed sprint planning should include the sprint goal and sprint backlog with full documentation of user stories as well as assignment to team members. A proper Kanban board tool should be used to track the status of user stories. A proper note taking tool should be used to record proceedings of Scrum ceremonies such as planning, daily stand-up, story time and retrospective. At the end of each sprint, evidence the deliverable of each user stories using a combination of source code and screenshots. Your group is expected to adopt a **full-code approach** for all project tasks.

For evidencing the Kanban board and meeting notes, your group may choose to take screenshots, provide public/guest link to the actual team workspace, or add me to your team's workspace. For the latter two options, please name your workspace to include your group number prominently. Link and account should use this email address distwk@nus.edu.sg. For source code repository, please prioritise the use of GitHub. Your group should name your repository to include your group number and add my GitHub username tanwk. Please use a **private** repository.

Please maintain your group's workspace and repository until the end of May 2026 to facilitate the evaluation process.

DataOps

For the purpose of this project, your group is required to undertake all relevant DataOps tasks leading to the delivery of the initial versions of the model. These tasks should include, but not limited to, dataset construction, ingestion, cleansing, transformation, watermarking, lineage tracking and exploratory data analysis. The DataOps tasks should be documented in the project report using a consolidated format. Scrum documentation is only required for one sprint.

It is mandatory to automate and orchestrate the DataOps processes with a suitable toolchain. For evidencing this requirement, please provide screenshots and/or source code.

MLOps

For the purpose of this project, your group is required to undertake all relevant MLOps tasks leading to the delivery of the initial versions of the model. These tasks should include, but not limited to, model development, preparation for production, deploying to production, monitoring and feedback loop and model governance. The MLOps tasks should be documented in the project report using a consolidated format. Scrum documentation is only required for one sprint.

It is mandatory to automate and orchestrate the MLOps processes with a suitable toolchain. For evidencing this requirement, please provide screenshots and/or source code.

DevOps

If your group is attempting the DevOps requirement, please ensure that a suitable business software application is conceptualised, designed and implemented. The software application should automate and integrate the analytics deliverables, especially the ML model, to derive tangible business values, e.g., capturing incremental revenue via an e-commerce product recommender system.

A scaled-down version of the software application with minimal user interface and functionalities is acceptable. For example, it is not necessary to have a complete set of e-commerce functionalities. You can focus on product catalogue browsing or shopping cart management to showcase the recommender system.

If attempting the DevOps requirement, it is also mandatory to automate and orchestrate the DevOps processes with a suitable toolchain. The DevOps process should also be synchronised with the MLOps processes, e.g., a new version of a model might necessitate changes to the software application. For evidencing this requirement, please provide screenshots and/or source code.

Use of Generative AI Tools

Your group is allowed to use generative AI tools for the project. However, you are not allowed to use a generative AI tool to outright generate the entire codebase of any single task. Such use of generative AI tools will be considered as an act of academic dishonesty and subjected to disciplinary actions.

You are only allowed to apply human-in-the-loop AI-assisted coding/analytics. Such use of generative AI tools must be documented in the written report with appropriate evidence. Failure to declare the use of generative AI tools or provide evidence will also be considered as an act of academic dishonesty.

Project Report Requirements

The **project report** is intended to document the work done and evidence the deliverable.

The written report must contain the following contents and formatted accordingly:

1. Cover page (include group number, and full name and student number of all group members).

2. Table of contents.

3. Formulation of Analytics Project:
   - Problem statement
   - Business value(s)
   - Solution statement

4. Agile Analytics Project Management with Scrum:
   - Overall project planning.
   - Detailed sprint planning for one sprint on DataOps
   - Detailed sprint planning for one sprint on MLOps

5. DataOps:
   - Dataset construction
   - Dataset ingestion, integration, cleansing, transformation, lineage, exploratory analysis, etc.
   - Automation and orchestration of DataOps process with suitable toolchain

6. MLOps:
   - Model development – Data exploration, feature engineering and selection, experimentation, model evaluation and comparison, version management and reproducibility, etc.
   - Preparing for production
   - Deploying to production
   - Monitoring and feedback loop
   - Model governance

7. DevOps:
   - Description of the software application.
   - List of implemented use cases (e.g., UML use case diagram).
   - Automation and integration with analytics deliverables and the associated business values realised.
   - Screenshots of prototype.

8. Group reflection on the use of Generative AI tools, including evidences of prompting, etc.

9. Conclusion

10. References (if any)

11. Appendices (if any)

Please format the written report based on the following settings:

- Paper size: A4
- Margins: 1 inch all around.
- Orientation: Portrait.
- Line Spacing: Double spacing.
- Font Type for Main Text: Times New Roman.
- Font Size for Main Text: 12 pt.
- Page Limit: There is no page limit.
- File Format: DOCX. Do not convert the document into PDF.
- File Name: Name the file as "**GPXX.docx**" where "GPXX" is your group number that will be given to you during registration. If your group number is "GP01", please name your document as "GP01.docx".

Project Presentation

The peer learning and sharing session should summarise the work done succinctly and evidence the deliverable in at most 12 minutes.

For the full demonstration video, please provide an in-depth walkthrough of all DataOps, MLOps and DevOps deliverables in at most 30 minutes. There is no need to repeat the oral presentation from the peer learning and sharing session. The video can be recorded with any tool and in any format.

All group members should be involved in the presentation and video recording.

Project Deliverable Submission

The final project deliverable to be submitted to the Canvas Assignment tool in Week 13 are to be placed in a single zip archive file with the following folder structure:

- **docs** subfolder containing:
  - Softcopy of a Microsoft Word document in DOCX format containing your group's project report
  - Softcopy of the slide deck used during the peer learning and sharing session.

- **source** subfolder containing:
  - All Python source files (.py and .ipynb) that constitute your deliverable.

- **data** subfolder containing:
  - All datasets used in the source files
  - Include raw datasets
  - Processed datasets

Upload this zip archive file to the designated Canvas Assignment: Assignments > Group Project - Deliverable. This is to be done by the group leader only. Name the file as "**GPXX.zip**" where "GPXX" is your group number that will be given to you during registration.

Your group's deliverables must be submitted latest by **Sunday, 19 April, 11:59 pm**. No project will be accepted for assessment after this date/time.

For the full demonstration video, please upload the video to the designated Canvas Assignment: Assignments > Group Project - Video. This is to be done by the group leader only. Name the file as "**GPXX.YYY**" where "GPXX" is your group number that will be given to you during registration and "YYY" is the file extension of the video format.

Your group's video must be submitted latest by **Tuesday, 21 April, 11:59 pm**. No video will be accepted for assessment after this date/time.

If the file size of any deliverables exceeds the upload limit of the Canvas Assignment tool, please submit it via cloud storage instead.

Peer Review Requirements

At the end of the semester in Week 13, each member is required to submit an individual peer review form. Feedback provided in the peer review will be taken seriously and you may not be given the team grade for the group project if you receive consistent negative review from your peers.

For the peer review form, please download the softcopy template from the Canvas Files tool folder: Assignments > Group Project. The submission deadline for the peer review form is **Sunday, 19 April, 11:59 pm**. The peer review form should be completed at the individual level and submitted via the Canvas Assignment: Assignments > Group Project – Peer Review. Name the file as "**GPXX_YYYYYYYYY.docx**" where "GPXX" is your group number that will be given to you during registration and "YYYYYYYYY" is your student number.

Plagiarism and Academic Honesty

The University takes a serious view against any forms of plagiarism. Please refer to the NUS Plagiarism Notice (http://www.comp.nus.edu.sg/cug/plagiarism) for more information.

Your group may use generative AI tools such as ChatGPT to assist you in the course of undertaking background research and the various project tasks. But you should not outright ask ChatGPT to prepare the deliverables for you. Anti-plagiarism and AI content detection tools may be employed to cross-check your group's final submission. If your group is suspected of committing an act of academic dishonesty, penalty may be imposed, and follow-up disciplinary action taken against you.

-- End of Group Project Specification --